

# Week 1

General Notation

Types of Machine Learning Models

Predictive Models

Probabilistic Models

Machine Learning Techniques

Supervised Learning

Regression

Classification

Unsupervised Learning

Dimensionality Reduction

Density Estimation

## General Notation

| Symbol                 | Meaning                                |
|------------------------|----------------------------------------|
| $\mathbb{R}$           | Real Numbers                           |
| $\mathbb{R}^+$         | Positive Reals                         |
| $\mathbb{R}^d$         | d-dimensional vector of reals          |
| $x$                    | vector                                 |
| $x_j$                  | $j^{th}$ coordinate                    |
| $\ x\ $                | Length of vector $x$                   |
| $x^1, x^2, \dots, x^n$ | Collection of $n$ vectors              |
| $x_j^i$                | $j^{th}$ coordinate of $i^{th}$ vector |
| $(x_1)^2$              | Square of 1st coordinate of vector $x$ |
| $1(2 \text{ is even})$ | 1                                      |
| $1(2 \text{ is odd})$  | 0                                      |

- $\text{Sign}(0) = +1$

## Types of Machine Learning Models

### Predictive Models

- Regression

- Classification

## Probabilistic Models

- Density Estimation

# Machine Learning Techniques

## Supervised Learning

### Regression

$$Loss = "$$

### Classification

$$Loss = "$$

## Unsupervised Learning

### Dimensionality Reduction

$$Loss = "$$

### Density Estimation

$$Loss = \frac{1}{n} \sum_{i=1}^n -\log(P(X^i))$$

During parameter estimation for a density estimation method such as Gaussian Mixture Model, negative log likelihood is minimized.